

**Retrieve all students:**

```
SELECT student_id, first_name, last_name, email
FROM student;
```

**Get student courses with grades and credits:**

```
SELECT
    e.student_id,
    c.course_name,
    c.credits,
    e.grade_letter,
    g.grade_point
FROM enrollment e
JOIN course c ON e.course_id = c.course_id
JOIN grade g ON e.grade_letter = g.grade_letter;
```

**Calculate GPA for a specific student:**

```
SELECT
    e.student_id,
    SUM(g.grade_point * c.credits) / SUM(c.credits) AS gpa
FROM enrollment e
JOIN course c ON e.course_id = c.course_id
JOIN grade g ON e.grade_letter = g.grade_letter
WHERE e.student_id = 'STU001'
GROUP BY e.student_id;
```

**Calculate required GPA to reach a target:**

```
SELECT
    e.student_id,
    3.50 AS target_gpa,

    SUM(g.grade_point * c.credits) AS current_points,

    SUM(c.credits) AS completed_credits,

    -- remaining credits (dynamic)
    ((SELECT total_credits FROM programme WHERE programme_id = 'P002')
    - SUM(c.credits)) AS remaining_credits,
    (
        (3.50 * (SELECT total_credits FROM programme WHERE programme_id = 'P002'))
        - SUM(g.grade_point * c.credits)
    )
```

```
/
(
  (SELECT total_credits FROM programme WHERE programme_id = 'P002')
  - SUM(c.credits)
) AS required_gpa

FROM enrollment e
JOIN course c ON e.course_id = c.course_id
JOIN grade g ON e.grade_letter = g.grade_letter
WHERE e.student_id = 'STU001'
GROUP BY e.student_id;
```